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THE FIRST VISIBILITY OF THE LUNAR CRESCENT

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ABSTRACT

The problem of predicting the moment when, after conjunction, the new crescent will become visible is both astronomical and physical. Although realized already by Ptolemy, the actual solution never did go much beyond the well-established Babylonian rule of thumb that the moon cannot be seen earlier than one day after conjunction.

This paper first discusses the importance of the sighting to the peoples of Islam and then mentions the criteria which control the phenomenon. This is followed by an outline of the theoretical solution given by the early Arab astronomers. It then proceeds to give a more accurate treatment, according to modern methods, which leads to rules by which the appearance and disappearance of the crescent can be predicted to within five minutes of time.

The second part of the paper presents translated extracts on the subject from the oldest sources, using modern astronomical nomenclature. These are taken from the Hindu compendia Surya Siddhanta and Pancha Siddhantika, and from al-Battani's Handbook of Astronomy.

1. INTRODUCTION

The Muslim year contains twelve lunar months, each starting at least one day after new moon, at sunset of the evening of the first sighting of the lunar crescent. The mean lunar month (the synodic month) has 29.53 mean solar days. If there are no clouds, the new crescent can always be seen 30 days after the previous one (a complete month), but in almost half of the cases it is seen already on the 29th day (an incomplete month). Therefore, no lunar month exceeds 30 days. Counting moons is believed to be one of the oldest ways of marking time. Nearly all primitive tribes watch for the new crescent and have ceremonies to mark the occasion.[1] Arabs greet the crescent with cries of "hilal", the ancient greeting of the arriving God, which later became the name of the new crescent.

For Muslims the most eagerly awaited, and therefore best observed, crescent is the one at the end of the month of Ramadan (the ninth month), which marks the end of the fasting period. People will gather on beaches or hills waiting for the faint sickle to appear in the pale blue edge of the yellow evening sky and, when it is suddenly sighted there is excitement and celebration. Following a tradition which al-Biruni[2] ascribes to al-Sakin: "When you observe the new moon of Rajab (the seventh month) count 59 days and then begin fasting", already on the evening of the 27th of Jamadi al-Akhir (the sixth month) people will be gathering on a hill to see if the moon of Rajab is visible. As soon as the crescent has been sighted by at

least two men of good repute, the party proceeds to the court or kadi to report the observation. The kadi may ask the men questions like how the horns of the crescent were pointed and how high the crescent was above the horizon. If this is in accord with the data obtained from astronomical tables he will put the sighting on record and forward the news to the capital. In ancient times people used to light fires on hills to spread the news of the sighting. In Islamic countries today, the actual sighting of the new crescent is announced that same evening by radio and television. The same procedure is followed for the crescent of the beginning of Sha'ban (the eighth month). After this, 30 days are added to the day of the crescent of Sha'ban and on this day the month of Ramadan and the Islamic fasting period is to start, that is if the crescent was not sighted on the previous day. The ultimate criterion in any case remains the actual physical sighting of the crescent. Fasting therefore may start or end one day sooner than officially ruled. Indeed, the Prophet says: "Fast when she appears, and cease fasting when she re-appears. But if the heavens are clouded, so as to prevent your sighting, count the month of Sha'ban as 30 days". Astronomical calculation is used only as an aid and a safeguard against mistakes but not to replace observation, a tradition which is said to be based on the saying of the Prophet that "we do not write, nor do we reckon".

The lunar month starts at sunset and the month of Ramadan is announced by cannonfire and illumination of the Minarets. During Ramadan fasting starts each morning "when you can distinguish a white thread from a black thread in the light of dawn" and it ends at sunset, "thereupon fast the entire day till the night" (night taken to begin at sunset). At the end of Ramadan the same observations determine the great Bayram feast from the first till the third of Shawwal (the tenth month).

2. THE ARCS OF SEPARATION, OF DESCENT AND OF LIGHT

The theoretical problem of first visibility of the lunar crescent is to predict from astronomical arguments the conditions under which the moon may be sighted for the first time after new moon. Accurate prediction will greatly facilitate the observation and, for this reason, much effort was made already by the early Islamic astronomers to master this problem theoretically. The moment of first sighting will necessarily occur after sunset following new moon. Before sunset the sky is too bright for the crescent to be seen. If the position of the moon after sunset is as drawn in Fig. 1, an important step in the solution of the problem is to find the conditions under which arc AB on the equator or the vertical arc DS will be minimal. Arc AB = a_S will be called *Arc of Separation* or Arcus Appartionis. Arc DS = a_D will be called *Arc of Descent* and is related to the Arc of Vision or Arcus Visionis. The minimum value of

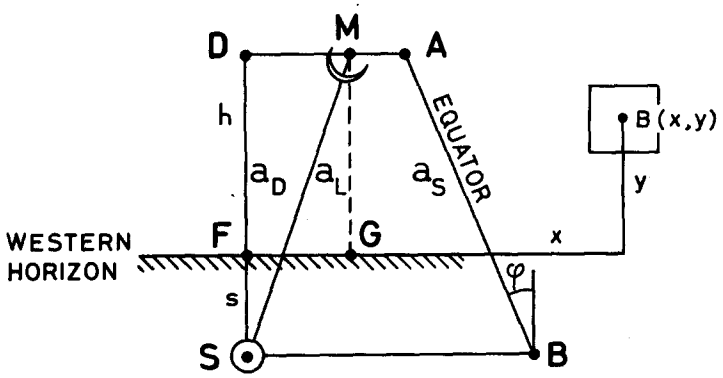


Fig. 1. Position of the moon after sunset with various celestial arcs of interest.

each depends on the brightness of the sky, on the brightness and shape of the crescent, on the local geographic latitude ϕ and on the difference in azimuth FG between the sun and the moon at sunset. The interrelation of these variables makes the visibility problem a difficult one and therefore a particular challenge to the theoretical astronomer. The situation is simplified by assuming that the solution does not depend on the position of the moon in azimuth, i.e. that we will find the same Arc of Separation for any position of the moon, as long as it has the same altitude. This idea was first put forward by Ptolemy and later used by al-Battani. Observations of the western sky after sunset show that in the region of interest this approximation holds well. For a given latitude the minimum Arc of Separation is now a fixed quantity. In Babylon it was taken to be 12 time-degrees or 4/5 th of an hour. Instead of 12 degrees in right ascension one can also give the minimum value of the Arc of Descent as a condition for visibility. If, again following Ptolemy, the western part of the celestial sphere is approximated by a plane and the circles on it by straight lines, the Arc of Descent is given by $a_D = a_S \cos \phi$. For Beirut this amounts to $12^\circ . \cos 34^\circ = 10^\circ$. In Fig. 1, a_D = DS = $h + s$, a_S = AB. In the theory of lunar visibility arc MS = a_L is also of importance. It is called the *Arc of Light*.

The problem of predicting the first crescent may have originated in Babylon and seems to have been transmitted to the Arabs by the Hindus. It was not given much attention in Greek astronomy. In the Khaqani zij of Ghiyath al-Din al-Kashi, we read (112 v) that a water-clock was used to measure to time of sunset. Since 4/5 th of an hour is the time in which a Hindu pinkan water-clock* sinks twice, the criterion for sighting the cresecent therefore was a time interval between sunset and moonset of two pinkans. In the critical situation the crescent was then seen only for a short moment at about one pinkan after sunset. These values of one and two pinkans seem to describe the ancient ritual of first sighting quite well, and were adopted—with further refinements—to the present day. For those who want only a simple rule of thumb we can phrase the Babylonian instruction as follows:

> The new crescent can *not* be seen less than one day after new moon. Therefore, look up in an ephemeris the time of new moon for the year and month concerned. Add 24 hours and see if this falls after sunset. If the moon sets later than 48 minutes after sunset the new crescent may become visible. In this case, about 20 minutes after sunset, begin watching the pale boundary between the blue sky and the orange glow of sunset. If the moon is not seen it will definitely be visible on the next evening, weather permitting.

Al-Sufi, in chapters 142 - 146 of his book on the portable planispheric astrolabe, describes how this instrument was used to find out whether the crescent can be seen or not. It is indeed a simple and straightforward method based on the Babylonian rule: The positions of the sun and the moon on the evening concerned are marked on the spider of the astrolabe. The sun is placed on the horizon of the tympan underneath the spider, and its position is read in time-degrees on the outer rim of the astrolabe. The spider is now turned till the mark of the moon is on the horizon and again the corresponding time-degrees are read off. If the sun and moon differ more than 12 time-degrees the crescent will be seen at about 6 time-degrees after sunset, otherwise not. Al-Sufi used his astrolabe in combination with a pinkan which had been supplied with a scale for this purpose.

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*A pinkan is a hemispherical bowl of brass or copper having a small orifice at the bottom. When placed on water it fills and sinks in 24 minutes.

Apart for the Babylonian data, the only extensive series of first and last visibility observations we have on record is that compiled in Athens between 1860 and 1967 by J. Schmidt.[3] These observations were discussed by Mommsen,[4] whereas Fotheringham[5] calculated for each crescent its azimuth and its Arc of Descent a_D at sunset. These, after correction for parallax, have been plotted in Fig. 2. In this diagram a cross means that the moon was seen and a circle that it was not seen. Unfortunately only three of these sightings (1, 2 and 3) are useful for the location of the critical lower limit. Nevertheless the graph shows that, as a rule, in Athens the moon cannot be seen below a_D = 9 degrees of lunar altitude at sunset, and that—always assuming a clear sky—sighting is fairly certain above 10 degrees. These values correspond to, respectively, 11 and 12 time-degrees for the Arc of Separation, in agreement with the Babylonian criterion which was also based on careful observation. We note that, although first sighting is indeed roughly independent of lunar azimuth, it is somewhat easier for large azimuth, where the Arc of Light is larger and the lunar sickle thicker.

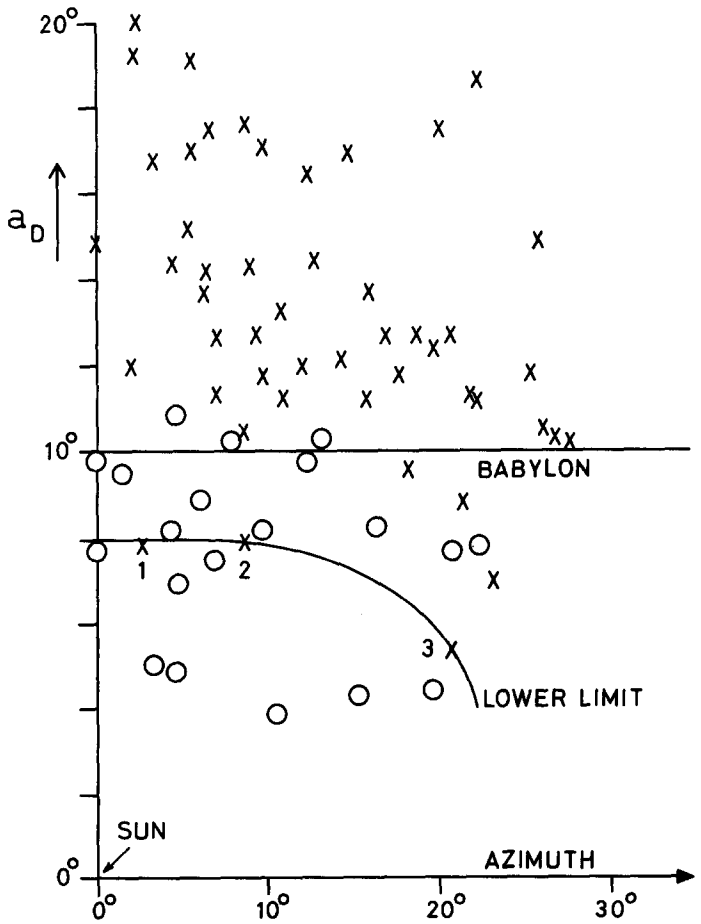


Fig. 2. Schmidt's observations according to Fotheringham.

In various places in the Islamic astronomical literature one finds mention of the conditions under which the new crescent may be seen. One finds limits and ranges for a_S, a_D and a_L and values without denomination. Also there are criteria of successive limitation which confine areas where the crescent may definitely be seen or not seen at all. Some of these criteria are depicted in Fig. 3. In all cases the sun is taken as a central reference. In this figure

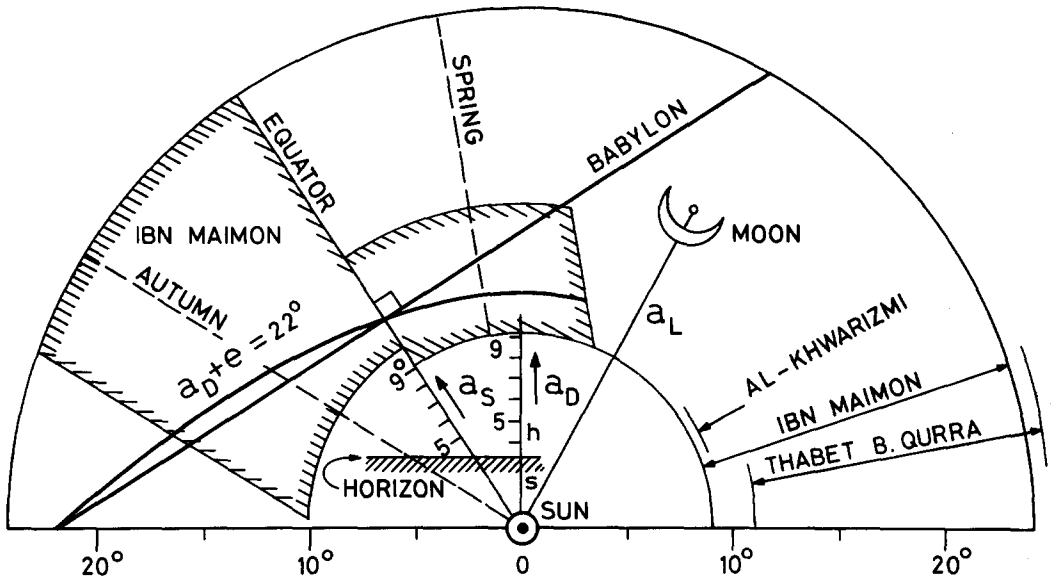


Fig. 3. Limits of visibility of the lunar crescent.

we note the slanting line for the minimum Arc of Separation of 12 degrees, used by the early astronomers (Babylonian, Hindu, Chinese). Then there is the upper and lower limit for the Arc of Light ($9^\circ < a_L < 24^\circ$) of al-Khwarizmi and Moses bin Maimon (Maimonides). When $a_L > 24^\circ$ the moon will always be seen, and when $a_L < 9^\circ$, the moon cannot be seen. Ibn Maimon, moreover, distinguishes between spring and autumn and places his limits according to the shaded areas in Fig. 3. For both regions he has a special condition, namely that $a_D + e \geqslant 22^\circ$, e being the so-called *elongation* between the sun and the moon. This condition is also drawn in the figure. For autumn it is the same as the Babylonian criterion, for spring the condition is less critical. Ibn Maimon states that the new crescent should be seen before the sun reaches a dip s below the horizon of 5°. This is in agreement with the older value of one pinkan (6 time-degrees) and fits well with our theoretical curve of Fig. 9, to be discussed below. Following al-Battani, various Arab treatises mention that the Arc of Separation should be taken smaller than 12° when the Arc of Light is large, i.e. when the moon has a large azimuth with respect to the sun. We will discuss this in Section 4.

3. ISLAMIC ASTRONOMICAL PROCEDURE

In order to predict the visibility of the new crescent one had to compute the Arc of Separation. Starting from a certain epoch at which the true positions of the sun and the moon are known, one had to find the respective apparent positions at sunset of the evening concerned. This was no simple matter because the motions of the sun and the moon show periodic variations and the moon may be on either side of the ecliptic with a varying latitude of up to 5 degrees. Also one had to take into account the lunar parallax, which near the horizon amounts to one degreee. Finally one had to relate the Arc of Separation on the equator to the solar and lunar ecliptic longitudes. Altogether, therefore, the computation of first visibility was a rather formidable problem involving nearly every aspect of mathematical astronomy. Ptolemy's planetary theory, however, already contains all the elements for handling a problem of this kind and the early Arab astronomers did not fail to make full use of this. Al-Khwarizimi[6] gives mathematical rules and tables for predicting the new crescent and al-Battani[7] presents a complete solution. Interesting in this respect also is the later account of ibn Maimon,[8,9] who largely follows al-Battani, but in addition discusses the legal aspects of the

sanctification of the lunar crescent. A good summary of the difficulties encountered is given by al-Biruni.[2] Since in the modern technique we follow a different approach, we will only give an outline of the ancient procedure.

In Ptolemaic astronomy the solar and lunar positions are given in ecliptic longitude and latitude. The Arc of Separation (AB in Fig. 4a) had to be related to these. This was done by computing arc HS in one way or another. In plane coordinates this arc is equal to

$$HS = HN + NS = \beta \tan (\phi + \delta) + e \qquad (1)$$

where e is the elongation or the difference in ecliptic longitude between the sun and the moon (being zero at new moon), β the lunar latitude and δ the angle between the ecliptic and the equator at the western horizon. Using the values for ecliptic longitude and latitude obtained from solar and lunar ephemeris tables, eqn (1) supplies the value of HS for the even ing of sighting. From Fig. 4a we also have

$$HS = \frac{a_D}{\cos (\phi + \delta)} \qquad (2)$$

from which follows

$$a_D = a_S \cos \phi \ , \qquad (3)$$

$$HS = \frac{a_S}{\cos \delta \tan \phi + \sin \delta} \ . \qquad (4)$$

In this equation, δ is a known function of the solar ecliptic longitude λ. If a_S = AB = 12°, for a given geographic latitude arc HS may be plotted or tabulated as a function of λ. This function, $R(\lambda, \phi)$ was presented in the so-called *Lunar Ripeness Table*. A graphical example of $R(\lambda)$ is shown in Fig. 4b. The moon would be visible if the value of HS from (1) would equal or exceed $R(\lambda)$. In the actual computation of $R(\lambda)$ the Islamic astronomer, in general, would not resort to our approximation of plane coordinates, but solve the problem exactly by spherical trigonometry.

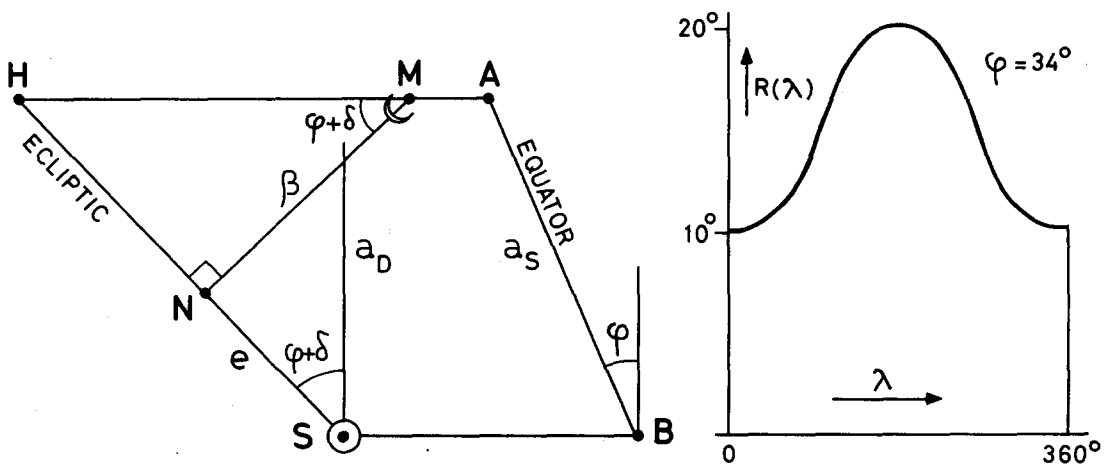


Fig. 4a, b. Construction of Lunar Ripeness Table.

4. THE WIDTH OF THE CRESCENT

Arabs, Hindus, Jews and, presumably, also the Babylonians, took the lunar and solar diameter to be 12 (or 15) digits and the width w of the crescent to increase linearly by 4/5 th of a

digit per day, to reach from zero at new moon—in 15 days—12 digits at full moon. Already in the early stages[10,14] it was realized that the width of the crescent is an important element in the criteria for visibility.

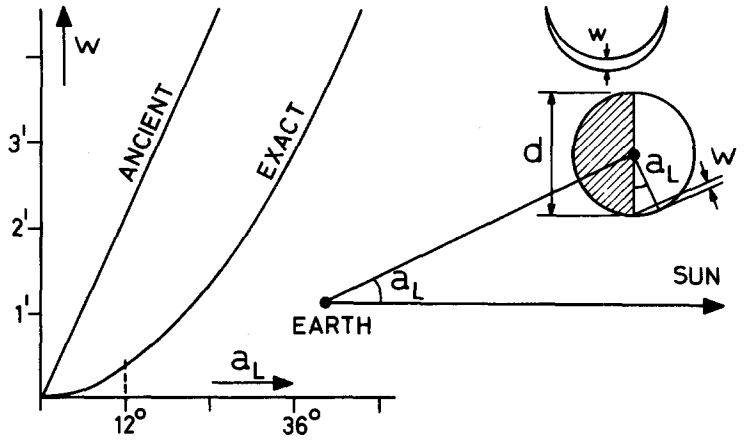


Fig. 5a, b. The width of the crescent

The width, as seen on a perfect lunar sphere, observed and illuminated from infinity, can be written down immediately using Fig. 5b:

$$w = \frac{d}{2}(1 - \cos a_L) = d \sin^2 \frac{a_L}{2} \quad . \tag{5}$$

In this expression, d is the lunar diameter and a_L the Arc of Light (equal to MS in Fig. 1). This result, taking $d = 30'$, together with the linear (Babylonian?) relation, is plotted in Fig. 5a. We see that, when $a_L = 12^\circ$, at about one day after new moon, the actual width is only 0.3', rather than the 2' of the ancients. This width is below the resolving power of the human eyes at sunset, so that in this case one would see the crescent a little less bright than it actually is. In most cases of factual sighting, however, the new crescent will have a larger width. For a given Arc of Separation, AB = 12° (Fig. 4), the moon's azimuth will depend on its elongation NS = e and latitude MN = β, and the Arc of Light MS = a_L may vary from about 10° (DS) to 20°. The Arc of Light will reach its maximum value at the time of the autumnal equinox, if the moon then also happens to have its maximum southward latitude of -5°. This situation is drawn in Fig. 6. The width of the crescent will then be about 1', three times its smallest value, and consequently the moon will be sighted more easily and earlier than on the assumption of an Arc of Separation of 12° only. Various Arab astronomers give rules for visibility which seem to account for a widening crescent. Whereas an Arc of Separation of 12° is taken for a narrow crescent, an arc as small as 10° may be accepted for a wide crescent, in agreement with the data of Fig. 2.

When the crescent is about 0.5' wide one may observe that the sickle is considerably less than half the circumference of the moon and that it is not smooth but irregular and grainy. Occasionally one will see that the illuminated sickle is interrupted in one or two places by black patches. The explanation of this is, of course, that the lunar mountains cast shadows which are particularly conspicuous for the new crescent. This phenomenon probably gave rise to the smiling profile of the "man in the moon" as depicted by artists. The author has found no evidence, however, of an early interpretation of the irregular contour as being due to mountains.

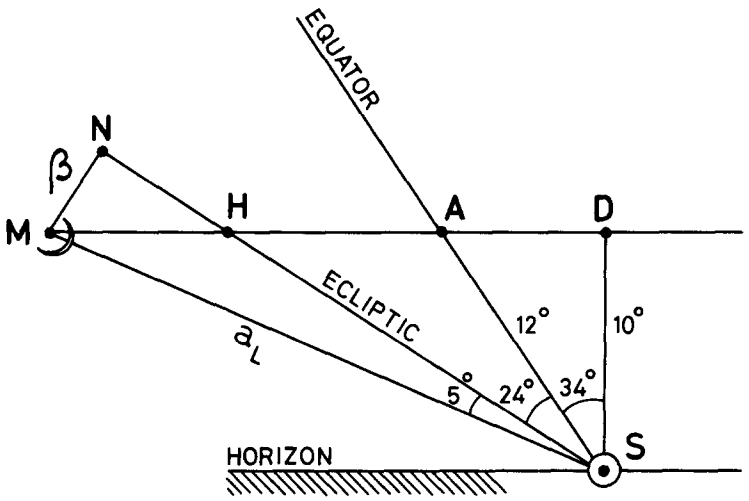


Fig. 6. The largest lunar azimuth.

5. MODERN ANALYSIS OF FIRST VISIBILITY

The object of this section is twofold: firstly we wish to replace the simple Babylonian rule of $a_S \geqslant 12^\circ$ by a more refined expression based on the relative brightness of the moon and the sky as well as on the measured ability of the human eye to observe contrast. Secondly we will express the solar and lunar positions in right ascension and declination rather than in ecliptic coordinates, because these are the ones found in a modern ephemeris.

Notwithstanding their most accurate tables, the Arab rules for crescent visibility are of a relatively simple and approximate kind, not going beyond an accuracy of half a degree. This is in agreement with the nature of the problem. The variable transparency of the sky and the uncertainty of the sighting refute a detailed theoretical approach. We will proceed along the same lines and present a graphical solution, without aiming at astronomical precision. In accordance with this, we make one more simplifying assumption about the brightness of the evening sky. That is, apart from taking the brightness to be independent of azimuth, we also assume that it does not depend on lunar altitude. This means that the brightness $B(x, y)$ of the sky above the western horizon after sunset at a certain moment is taken to be the same at every point (x, y). Measurements show that this is fairly correct at any time after sunset for

$$-30^\circ < x < +30^\circ$$

$$+1^\circ < y < +20^\circ \,. \qquad \text{(See Fig. 1.)}$$

Thus our theory will be based on the assumption of a western sky of homogeneous brightness, decreasing uniformly at every point as the sun descends below the horizon.

The average brightness of the sky can be measured directly with an ordinary photographic exposure meter. The author's results, obtained in this way, were the same as those collected in Maryland in 1952 with more specialized equipment and more accurate measurements by Kooman *et al.*[11] The combined results are plotted in Fig. 7. The brightness of the moon as a function of altitude (Fig. 8) follows the extinction curve of Bemporad.[12] The maximum brightness of the moon in the zenith at night, as observed from sea level, was determined with respect to the brightness of the sky by means of a series of photographs of different exposure on the

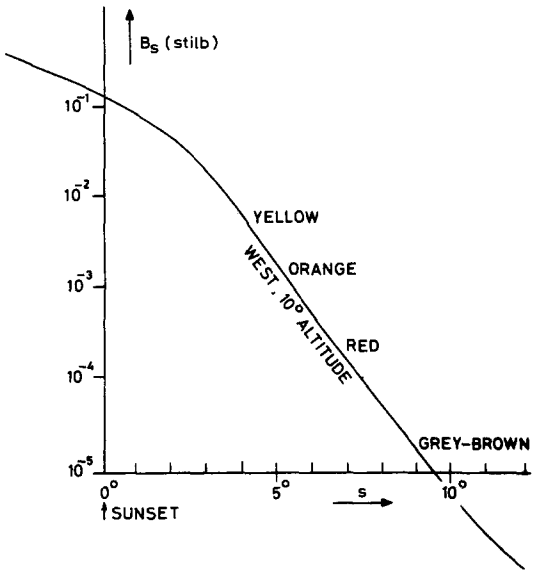


Fig. 7. Mean brightness B_s of the western sky after sunset, as a function of the solar dip angle s.

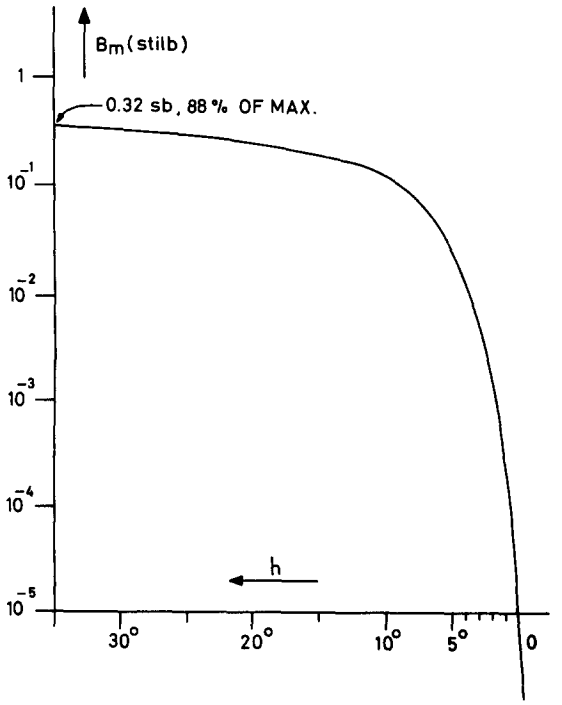


Fig. 8. Brightness B_m of the full moon at night, as a function of the altitude h in degrees.

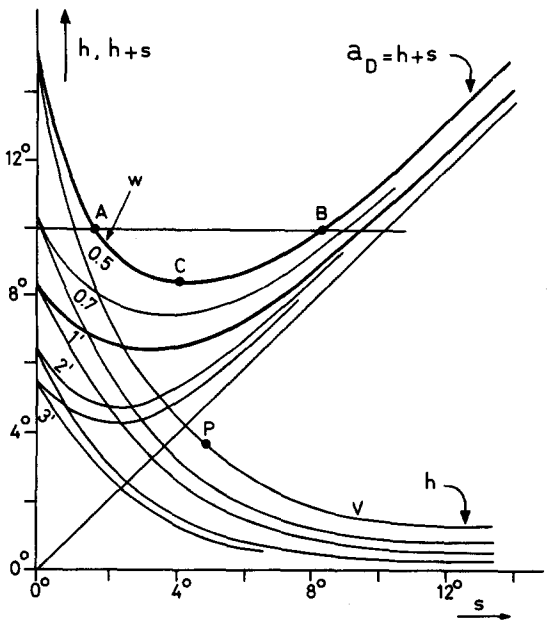


Fig. 9. Lunar visibility curves.

same film and was found to be 0.36 stilb. The brightness of the moon was taken to be uniform over its surface, although the slanting light from the sun on the early crescent would cause it there to be somewhat brighter.

In order to find whether the new crescent is visible we make use of the data for the minimum contrast observable by the human eye as given by Siedentopf,[13] and ignore the effect of colour. These curves are for circular disks and that for a diameter of d = 30' could serve for the full moon. We may apply it, as a first check, to find out whether the full moon can be seen in daylight: the brightness of the Mediterranean sky at noon is about 1 stilb. For this case Siedentopf's diagram of minimum visibility informs us that an object of 30' diameter requires a contrast of $C = -2$ in order to be seen. The brightness of the moon in daytime is the sum of the lunar brightness as observed at night plus the brightness of the sky which happens to be in front of the moon. Therefore the lunar brightness need only be 1% that of the sky, or 10^{-2} stilb. According to Fig. 8 the moon reaches this brightness at an altitude of 3°. We find, therefore, that according to these data the moon, when almost full, can be seen in daylight at any position higher than 3° above the horizon. Since this is confirmed by observation, the above approach seems to be sound.

Now the young crescent is, of course, not a circular disk but a sickle which, when one day old, has a width at its middle of only 0.3 minutes of arc. In order to still make use of Siedentopf's diagram we make the assumption that the visibility of a crescent of that width will be about equivalent to a circular disk of 0.3' diameter. Eventual discrepancies will then later be accounted for by some kind of "Gestalt" factor. Accepting this, we proceed as follows to construct Fig. 9. We assume a certain brightness of the sky, say 10^{-2} stilb and find from Fig. 7 the corresponding solar dip ($s = 4^{\circ}$). Next, from Siedentopf's diagram, we find that the required minimum contrast of an object of say, 0.5' diameter, in order to see it against a background of $B_b = 10^{-2}$ stilb, equals $C = 0$, or that $P_o/P_b = 2$. This shows that in this case the lunar brightness should also be 10^{-2} stilb. Bemporad's extinction curve (Fig. 8) now gives the corresponding lunar altitude: $h = 5^{\circ}$. This enables us to construct point P in the diagram of Fig. 9. Repeating this procedure the entire curve v can be plotted. For each value of the solar dip this curve gives the minimum altitude h at which a crescent of 0.5' width can be seen. In Fig. 9 similar curves have been constructed for a crescent of width $w = 0.7'$, $w = 1'$, $w = 2'$ and $w = 3'$, of which only those between 0.5' and 1' are of practical value. In order to make the result more convenient for use we notice that, while the sun and the moon are setting, their Arc of Descent $a_D = h + s$ remains practically constant. Therefore, apart from $h = f(s)$, in Fig. 9 we have also plotted $a_D = h + s = f(s)$. These latter curves can be constructed directly from the former by adding s to every ordinate h. We are now ready to apply these results.

Suppose, by way of example, that on a certain evening after new moon, $s + h = 10^{\circ}$. In the $(h + s) = f(s)$ diagram, $s + h = 10^{\circ}$ is a horizontal line which intersects the minimum visibility curve in points A and B. The crescent will be visible for the first time at point A and will remain visible for six degrees of solar dip (about 30 minutes), until point B. For $s + h = 9^{\circ}$ the minimum C of the curve will be reached, in which case the crescent would be visible for one moment only, namely when $s = 4^{\circ}$ and $h = 5^{\circ}$. This is the optimum situation. The visibility curves of Fig. 9 replace the Babylonian rule that $a_D = s + h$ be 10°. Apart from instructing us *when* the crescent will be visible, they also tell us *how long* it will be visible.

6. HOW TO PREDICT THE SIGHTING OF THE NEW CRESCENT

We now present the instructions by which the position and the time of sighting of the lunar crescent can be found. For following these an understanding of the preceeding theory is *not* required.

In the (American) Ephemeris one looks up the table for the sun (for 0^h ephemeris time) and writes down the true right ascension $\alpha_\odot$ and inclination $\delta_\odot$ of the sun for the day *after* the day of lunar sighting. The day *after* is chosen because these are the data for midnight which is only 6 hours after sunset.

Next one writes down the true right ascension $\alpha_\mathbb{C}$ and declination $\delta_\mathbb{C}$ of the moon for 18^h local time of the day of sighting. Now $\delta_\odot$, $a_S = (\alpha_\mathbb{C} - \alpha_\odot)$, and $\delta_\mathbb{C}$ are plotted in a diagram, starting from the sun S, as illustrated in Fig. 10.

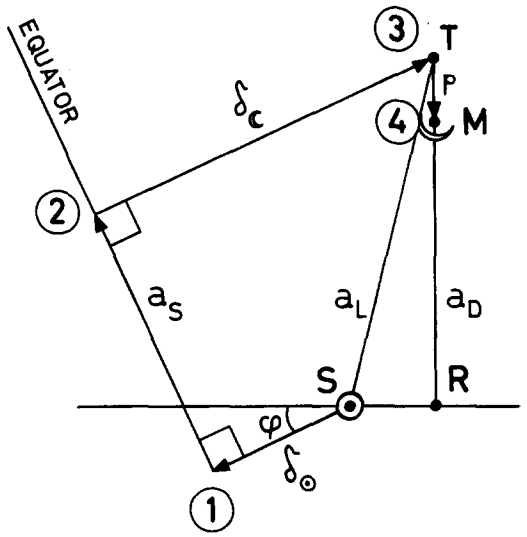


Fig. 10. Construction for finding the crescent.

Finally the terminal point T, representing the true position of the moon, is lowered vertically by one degree in order to account for lunar parallax. M is the position at which the moon will be seen. The diagram now contains the basic information and represents, in fact, a picture of the western sky at the moment of sunset.

TR = $a_D = s + h$ is the Arc of Descent, while TS = a_L is the Arc of Light. The latter, together with Fig. 5a, gives us the width of the crescent, which is required to select the suitable curve in Fig. 9. The former ($a_D = s + h$) can be marked off in the same figure against the selected curve to give s of the sun and h of the moon. Through this the problem is solved. The corresponding differences in time can be read on the equator of Fig. 10, taking one degree equal to four minutes of time. In order to obtain absolute times, one also needs the time of sunset, which can be obtained from the observation of previous sunsets.

In the past ten years careful observations by the author of the new crescent revealed that the present theory permits the prediction of visibility, for an evening when the sky is clear, to within five minutes of time. No correction was found to be necessary for our assumption that a sickle of width w is equivalent to a circular spot of diameter w. It would seem that further refinement of the above theory would serve no practical purpose.

7. THE HINDUS ON FIRST VISIBILITY

It is fairly certain that the early Muslim astronomers extracted most of their knowledge of the lunar motion and the crescent from Indian books dating roughly from the year 500 A.D. Since the sources on which the Hindus have drawn are still unknown, these books form the

oldest source of information on lunar sighting. Greek astronomical works refer only scantily to the subject. That the lunar month begins with the new crescent is mentioned by Aratos (around 300 B.C.):

"Look! When the moon in the West shows very thin horns
She tells us that the month begins".

That the crescent can be seen in the second day after conjunction is only implied by Aratos, but stated explicitly by his contemporary Berossos the Chaldean (as quoted by Vitruvius):

"...The outer edge of its luminous hemisphere seen on earth as a very thin arc announces the second lunar (day)".

Aratos also discusses the significance of the tilting of the horns, relating the phenomenon to the forthcoming weather. In similar fashion we find it mentioned in Ptolemy's Tetrabiblos. The construction of the crescent by means of two circles is discussed by Cleomedes (*ca.* 50 B.C.), but this author mentions nothing about lunar sighting or any theory about it. The Hindu books are therefore the basic material from which to start.

In the following we present the fifteen stanzas on lunar crescent visibility of Chapter 10 of the Surya-Siddhanta and the corresponding ten stanzas of the Pancha-Siddhantika. In these translations we deviate from the usual practice of giving a very accurate but obscure word-by-word translation accompanied by a detailed explanation in footnotes and fine print. Such material is long since available.[10,15] Instead the translation is in ordinary English and conventional uniform notation, so that the astronomical contents can be understood immediately. Nevertheless we have made every effort to follow the original text as closely as possible. What is not found in the original is placed between brackets, including formulae and figures.

Surya-Siddhanta

Chapter 10

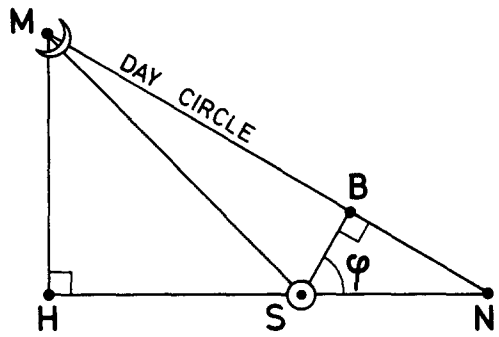
The rising and setting of the moon and the tilting of her horns

1. The calculation of the heliacal rising and setting of the moon also is to be made by the rules already given. At an Arc of Separation of two nadis* she becomes visible in the West, or invisible in the East.
2. Add six (zodiacal) Signs to the longitude of the sun and moon respectively and find, as before, the Arc of Separation and its corresponding rising in respiration. (However), if the sun and moon are in the same Sign find their elongation in minutes (of arc only).
3. Multiply the (first) result, in nadis, (respectively) by the daily progression of the sun and of the moon divided by sixty. Add the correction due to each motion thus found to the longitude of each, and (from this) find again their Arc of Separation in respirations.
4. Continue this until the interval, in respirations, of the sun and the moon becomes fixed. By so many respirations the moon will go to her setting after the sun during the waxing first half of the month.
5. Add half an orbit to the sun's longitude, and calculate the corresponding Arc of Separation in respirations. By so many respirations after sunset the moon will reach her rising in the waning (second) half of the month.

*One nadi equals 24 minutes of time, or 6 equatorial degrees. One respiration equals 4 seconds of time, or one equatorial minute of arc.

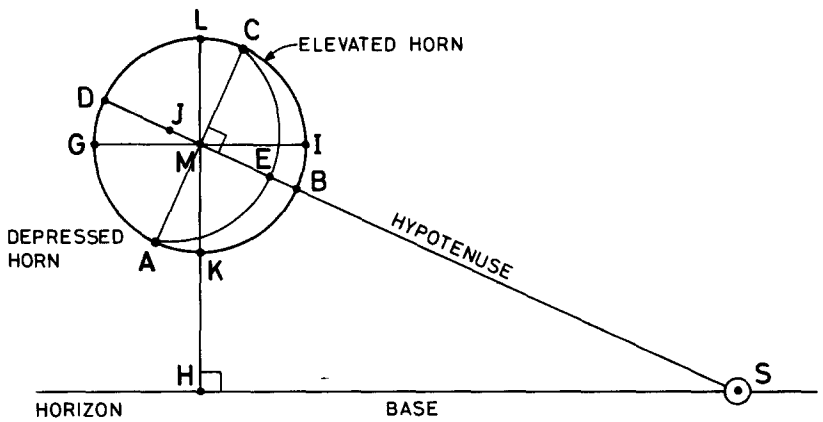
(Tilting of the Horns)

6. Take (Fig. 11) the difference (SB) of the solar and lunar declinations, if their directions are the same. Take the sum in the contrary case. The sine of this (difference) is to be regarded as South or North, with respect to the direction from the sun to the moon.



(Fig. 11.)

7. Multiply this (sine) by the hypotenuse of the moon's (equinoctial) shadow at sunset.* (The result is BN). If (B) is North, subtract it from the sine of the distance (on the day-circle,BM), and multiply by 12. If (B) is South, add it to the same.
8. The result (MN), multiplied by the sine of the (geographic) latitude, gives the base (HN), in its own direction. The gnomon is the perpendicular (MH). The square root of the sum of their squares is the hypotenuse.
9. The number of minutes (of arc) in the longitude of the moon diminished by that of the sun gives, when divided by 900,** her illuminated part (in Greek digits). This, multiplied by the number of (Hindu) digits of the moon's disk (fifteen), divided by 12, gives the corrected value of the same.



(Fig. 12.)

*The Sanskrit text has "midday" instead of "sunset". With our correction the moon's shadow equals the tangent of geographic latitude.

**The full moon at half its orbit or 180°, or 180 × 60' = 10,800', has 12 digits of light. Therefore 1 digit corresponds to 10,800'/12 = 900'.

10. Take point S (as centre) of the sun (Fig. 12). From it lay off the base (SH) in its proper direction. Also (lay off) the hypotenuse, passing through the extremeity (M) of the perpendicular (MH) and the central point (S).
11. Around the point of intersection (M) of the perpendicular and the hypotenuse describe the moon's disk, according to its size at the given time. Then, by means of the hypotenuse, first ascertain the (lunar) directions (BD and AC).
12. Lay off on the hypotenuse, from its intersection with the disk (B), in an inward direction, the measure (BE) of the illuminated part. Between the terminator (E) of the illuminated part and the North and South points (C and A) draw two fish figures.
13. From the point of intersection (J), of the lines passing through their midsts, describe an arc passing through the three points (A, E, C). With the disk already drawn, such will be the moon on that day.
14. After ascertaining the (cardinal) directions (KL and GI) by means of the perpendicular (HM), mark the elevated horn (C) at the extremity of the cross-line (AC). Having made the perpendicular to be vertical, this is how the moon will appear.
15. For the waning half of the month subtract the longitude of the sun increased by six Signs from that of the moon and calculate, in the same manner as before, her darkening part. In this case lay off the base and the circle of the moon in the opposite direction towards the North (the text has: West).

Pancha Siddhantika of Varaha Mihira

Chapter 5

On the appearance of the moon

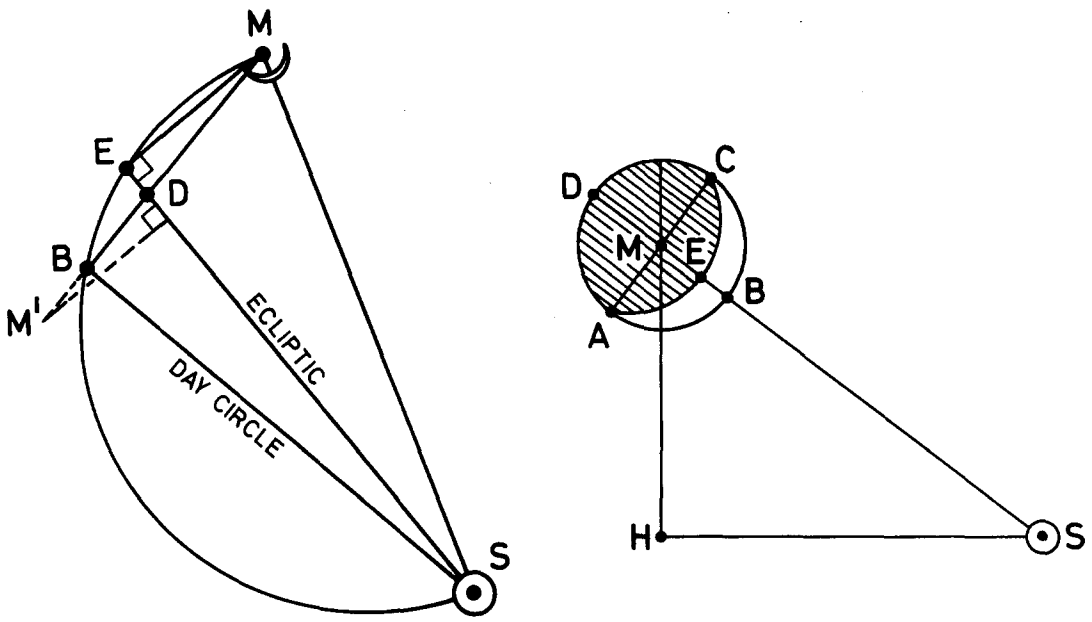
1. Take (Fig. 13a) the sum (MS + MB) and the difference (MS - MB) of the distance between sun and moon (MS) as well as the difference of their declinations (MB). Multiply the two quantitites thus obtained:

$$(MS + MB)\ (MS - MB) = (MS^2 - MB^2) = (BS)^2$$

and take the square root of what results. Divide the difference (BD) of the declinations by this result (BS), after having multiplied it by the latitude (EM) of the moon:

$$ED = \frac{EM \cdot BD}{BS} \quad .$$

2. The result (ED) has to be deducted from the elongation (ES) between the sun and the moon, if the lunar latitude (EM) has the same direction as the difference of the declinations (BD). Otherwise the result (ED) has to be added. In the case of morning twilight (when the moon will disappear) the instructions for adding and subtracting are to be reversed.
3. If this (adjusted elongation ES ± ED) is laid off on the ecliptic, from (a point in the) seventh Sign (opposite the sun) and if (this arc) rises in two nadis, then—if the sky is clear—the moon is seen.
4. (The number of digits of the full moon is 15 (and since 180'/15 = 12') the 12th part of the hypotenuse (MS) represents the illuminated part (BE) of the moon. It is to be marked off in the direction of the diameter (BD)starting from B. The figure is the same as Fig. 12).
5. (For this paragraph see Fig. 11.) The perpendicular (MH) is obtained by (first taking) the difference of the declinations (SB and the distance (BM on the day-circle F.). Of these two quantities (BN and BM) take the sum if their direction is the same, otherwise (take) the difference. (Multiply the result with the cosine of the geographic latitude. What you get is the perpendicular.) The hypotenuse (MS) is the distance between sun and moon. The square root of the difference of the squares of these two sides is the base (HS).



(Fig. 13a, b.)

6. Let the perpendicular (MH, Fig. 13b) be graduated in the direction in which the sun is with regard to the moon, one division being taken equal to one digit. The base (HS) and the hypotenuse (MS) are to be graduated similarly.
7. First graduate the hypotenuse starting from the center (M) of the moon. Then, from the hypotenuse (graduate) the perpendicular. Then the diameter (BD) in the direction of the lunar centre. This diameter on the circumference of the moon is called "director". The illuminated part is marked off from the middle (B). It has the shape of a bow.
8. Multiply (the moon's) southern or northern latitude by the equinoctial shadow divided by 12.* If the moon is rising, the resulting degrees are to be added (if the latitude is southern), and to be deducted (if the latitude is northern). The reverse has to be done in the case of setting.
9. Subtract the (longitude) of the moon from the longitude of the sun (and find the corresponding number of Signs between them). Depending on whether these are less or more than six, the rising of the moon has to be determined as taking place during the day or during the night, according to the rising of the sun.
10. Having thus (i.e. in the way indicated in stanza 8) performed the subtraction or addition, deduct the difference of lunar and solar longitude from half the orbit. In the time (reckoned from sunrise) equal to the time of the rising of the remaining Signs the moon goes to her setting.

8. AL-BATTANI ON THE LUNAR CRESCENT

The first Islamic astronomers who wrote on the appearance of the lunar crescent are those of the Abbasid court in Baghdad. Best known among these are Habash (740 - 840), al-Khwarizmi (830), al-Farghani (850) and al-Battani (850 - 929). Al-Biruni[2] in his "Chronology" writes about the moon:

*12 being the length of the gnomon, and the equivalent equinoctial shadow multiplied by the tangent of the geographic latitude 12 equals the tangent of the geographic latitude.

"The computation of the appearance of the new crescent is a very long and difficult procedure, the demonstration of which requires long calculations and many tables. What is needed for this one can read in the Handbook of Astronomy, written by Muhammad ibn Gabir al-Battani, or in one of the works written by the mathematician Habash."

The Handbook of Astronomy of Al-Battani was edited in 1903 by C. Nallino,[7] together with a Latin translation and commentary. Chapter 41 of this Handbook deals with the new crescent. In the following we present a new translation from the Arabic into English which has been carefully compared with the Latin of Nallino. This new version, based on the independent translations by Faruk Adman and Abdul-Rahin Jauni, is believed to follow the original text more accurately, although it rarely deviates from Nallino in its astronomical meaning. Since the Latin commentary by the Italian astronomer Schiaparelli is also of some historical interest, we have added an English translation of this as well.

Al-Battani

Handbook of Astronomy

Chapter 41

On seeing the crescent at the beginnings and ends of the months. On the azimuth of its position where it can be seen, being high or low. On its shape as related to the brightness of its light. On the straightness of its horns and their inclination with respect to the ecliptic.

(1. The ancient opinion and the Arc of Separation)

Finding the way of seeing the moon at the beginnings and ends of the months is one of the useful products of science, because the time-reckoning of the Arabs and the beginnings of the months depend on seeing the crescent. Its knowledge is in many ways a truly difficult matter: (One has to know) how far the moon is from the sun, how close or how far from the earth, the variation in (ecliptic) latitude of the moon towards the North or the South, the parallax of the moon in longitude and latitude in each country, the duration of the rising and setting of the zodiacal Signs in the (various) climates and (finally) the variation in its brightness.

So, of the men of our time, some who wanted to study the way of seeing the crescent committed errors and failed to achieve the truth of the matter. They thought that the declination of this luminary and its latitutde are the same arc. They assumed that the parallax of the moon is not due to variation in altitude but to the distance in zodiacal degrees of the moon from the prime meridian. (Finally) they confused arcs with (the linear) sines obtained from earlier sources, (leading to results) which could neither be fitted to measurement nor proved to be true.

For the ancients there was no need to know (all of) this, because they had their records from which they constructed the solar years and because they determined the beginnings of the lunar months by the times of conjunction which can be found exactly by calculation. They stated that *the crescent can not be seen in less than one day and one night (after new moon)* and, if an investigation is made for the causes of the appearance, it is found that this statement is (indeed) the basis from which one should start. The Arc of Separation found through observation is very nearly equal to the mentioned amount. However, since there are also perceptible differences, it is clear that (the phenomenon) had not been understood completely, but only approximately.

The conclusion whether the crescent can actually be seen can be reached correctly only by starting from the equator arcs between the (parallels of the) sun and the moon at sunrise or sunset. If these arcs are observed in one of the Climates, then the Arc of Separation is fixed for all Climates. Everyone agrees that this Arc of Separation by observation is found to be about 12 time-degrees.*

It was (already made) clear, that the moon's (mean) motion away from the sun is $12^{o}11'$ per day and night, if the sun's deviation from its mean motion is ignored. This is the (daily increase in) elongation between the sun and the moon in ecliptic degrees and is approximately the same as has been observed for the equator degrees. An amount (of 12 time-degrees) equals 4/5 th of an hour. The moon is going away from the sun by an amount of 2/5 th of a degree per mean hour. It follows that, when the sun sets and (at that moment) between it and the moon are approximately 11 3/4 time-degrees, the moon will not set before $12^{o}11'$ have been completed. Therefore, in this respect, the mean Arc of Separation will be 11 3/4 time-degrees, which is the arc of rising and setting of the (corresponding) zodiacal Signs in the countries (concerned).

If the distance of the moon from the sun equals this amount in ecliptic degrees,* the illuminated part of the moon will be about 4/5 th of a digit, the whole disk of the moon being 12 digits. At the time of appearance, the moon may be away from the sun more or less than this amount. Its light increases or decreases according to the distance and, consequently, it may appear at a larger or smaller arc. At the same time the moon may be closer to—or farther away from—the earth, depending on its position in the epicycle, and this will (also give rise to) a waxing or waning of these amounts. As a result the crescent will not appear according to one single arc, but to many different arcs.

(2. Computation of the Corrected Elongation*)

So, (if you wish to determine in this way) whether the new crescent can be seen or not, you should determine the position of the sun and the moon at sunset for the day after conjunction, which is on the 29th day of the Arabic month. Determine their true positions in the ecliptic for the country you wish, and determine the true latitude of the moon and its sign (+ or -). Then determine the parallax of the moon in longitude and latitude at the time of sunset at this location, so that you obtain the apparent position of the moon in ecliptic longitude and latitude, and the sign of the latitude. After this, find the apparent equatorial distance from it (the moon at sunset) to the degree at mid-sky. Through this is known half of the arc of a lunar day, that is, half (of the arc) in which its stays above the earth. (This is) as was explained earlier in the book, in the chapter (28) concerned with the declination of stars and their (equator) degree at mid-sky, according to the latitude of the star and the declination of the degree (of the equator) at which it is. It is (also found) in Chapter (19) on half the daily arc of stars of given delination.

What is found for the moon's half daily arc is added to the right ascension of its degree of mid-sky. What is obtained is the (time of) rising of the degree opposite the one in which the moon will set in that climate. What remains is the Arc of Separation in time-degrees between the sun and the moon at sunset. Keep this in mind.

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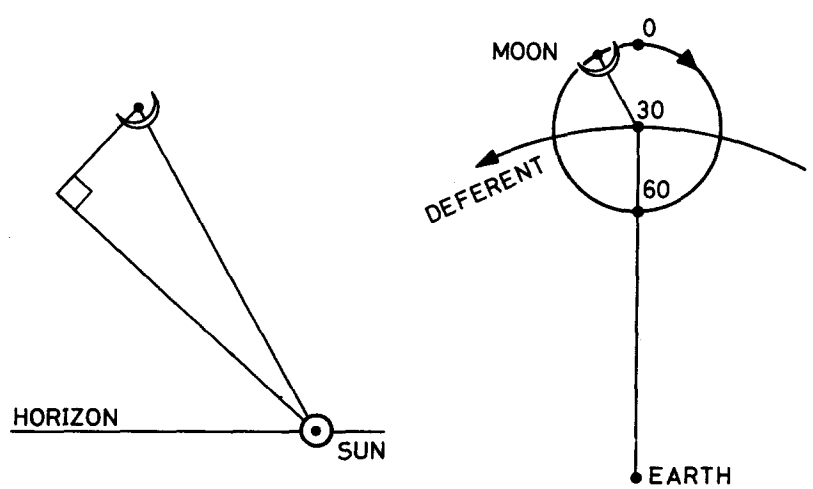
*Al-Battani writes "Arc of Vision". It is actually what we call elongation, but since al-Battani assumes the sun and moon to be in the equator, these two are the same. This assumption represents a kind of average situation.

Now find the true degree (of longitude) of the moon and its true latitude, take the difference of the (ecliptic) degrees of the true sun and moon and multiply this by itself (Fig. 14a). Then add it to the latitude of the moon, after also multiplying it by its (own) amount. Take the square root of the sum. What results is approximately the distance between the sun and the moon. (The same result can be obtained) more correctly and accurately from what we mentioned earlier in the book, in the Chapter (26) on the determination of distances between stars on the celestial sphere.

If the elongation between the sun and the moon differs from $12^{\circ}11'$, you find how much it is more or less than $12^{\circ}11'$, and determine by means of a drawing what fraction the increase or decrease is (of $12^{\circ}11'$), which is then the fraction of the apparent moonlight. Take this result, whether it be increase or decrease, and express it in fingers. This (fraction) is the increase or decrease of the Arc of Light.

(3. Correction for variation in lunar distance from the earth)

Next look up in the Calendar Table the corrected anomaly of the moon, and take the corresponding parts in the third column (of Table T II) which contains the parts of the lunar distance (Fig. 14b). Suppose these are 30 parts. In this case the moon is at its mean distance from the earth, and, if the fraction were a decrease, you add it to $11^{\circ}45'$. In a case where the parts are more or less than 30, you see how much more or less it is, and determine the corresponding ratio. Take 1/12 th of a result, (because) if the diameter of the moon changes, its increase or decrease will be about 1/12 th of its mean diameter. This 1/12 th is added to the fraction if the fraction was an increase, and if the parts of the third column are more than 30. If the parts of the third column are less than 30, you subtract the 1/12 th part from the fraction. If (however) the fraction is a decrease and the parts are more than 30, you subtract the 1/12 th part you got from the fraction. If the parts are less than 30, add it to the fraction. If the fraction that results after addition or subtraction is more than $12^{\circ}11'$, it is subtracted from $11^{\circ}45'$. If it is less, you add it to $11^{\circ}45'$. What you obtain is the amount of the corrected elongation, corrected for the increase or decrease of the moon's light according to its distance from the earth.



(Fig. 14a, b.)

If the Arc of Separation in time-degrees between the setting of the sun and the moon, which was kept in mind, is equal to or more than the corrected elongation, the crescent will be seen. If it is less than the corrected elongation it cannot be seen in that country.

Clear and pure weather is required (in order to see the crescent). A thick and opaque (sky) makes it difficult to do so, because, at the time of sighting, it causes the image to vary from intense to weak. During early twilight the atmosphere is bright; after that, before the moon disappears near the horizon, it becomes less intense. The crescent can be seen after the moment of appearance established (by computation) and before approaching the phase of disappearance. It is, therefore, necessary not to lose hope of seeing the crescent before it is certain that it has disappeared. Only when the moon has passed the region of visibility and it is certain that it has gone down beyond the horizon, has it become hopeless. For these reasons the moon may be seen in one location and not in another. This may also happen because of the variable duration of rising and setting of the zodiacal Signs in different countries.

(4. Limits of the Arc of Vision)

(Let us now return to) the opinion which the ancients had reached from observation of the crescent, about the truth of which there is no doubt. We mean their stated and recorded fact that (the moon) can not be seen less than one day and night (after conjunction). Let us determine the distance between the moon and the sun when the moon travels at its lowest speed and the sun at its highest. This (is the case) when the moon is farthest (from the earth) and the sun is closest. We (then) find that in one day and night the (moon's) elongation from the sun is (only) 10 5/6 th ecliptic degrees. If, (on the other hand), the moon travels at its highest speed and the sun at its lowest, i.e. when the moon is in the position closest (to the earth) and the sun at its remotest distance from the circular orbit, we find that in one day and night the moon's elongation from the sun is about 13 2/3 degrees. These amounts of ecliptic (degrees) we use to find the Arc of Separation of the moon at the time of observation. We say that, when between the sun and the moon there are either 10 5/6 ecliptic degrees or 13 2/3 ecliptic degrees, (the moon) is in a position where it can be seen, (i.e.) if there is nothing to obstruct this, such as the weather conditions we mentioned. This we can say right away (for the extreme cases). But the moon may be away from the sun by more or less than these ecliptic degrees, because in its epicycle it may have moved from the point of farthest distance to a closer position. Because of this, as we mentioned before, the amount of the Arc of Separation may vary.

(If you wish to know) the conditions of appearance, look up (,in a table, the coordinates) for the moon and the sun for the mentioned time in the way we have shown, in order to find the moon's elongation from the sun in degrees at sunset according to (the latitude of) the country. Thus you find the distance of the moon from the sun from the ecliptic degrees and the moon's latitude (The Arc of Light). If (the distance) is larger than $13^{\circ}40'$, you determine the increase. If it is less, you find the decrease. You see what fraction one of these is of $13^{\circ}40'$, subtract a similar amount, and this is the fraction.

Suppose that, at the assigned time of the given elongation, the moon is at its farthest distance (from the earth). This happens when the anomaly of the moon is about 360°—and not much more or less than this, except for an imperceptible amount. In this case subtract that fraction from $10^{\circ}50'$ if it is an increase and add it to $10^{\circ}50'$ if it is a decrease. What you get is the corrected elongation.

If the moon has passed its remotest point, you look up the corrected anomaly of the moon in the column of the table, and take the parts of the third column and their fraction of 60. You take an equal amount of the ratio (which results), then take 1/5 th of what you have got. (This is done) because 1/5 th of the largest diameter of the moon equals the difference between

the largest and smallest diameter of the moon. So you take 1/5 th (part) and subtract it from the ratio you got if the fraction is a decrease, and add it if the fraction is an increase. What results, after (considering) the increase or decrease, you express in equator degrees. After this conversion you add it to $13^{o}40'$ if it were a decrease, and you subtract it from $13^{o}40'$ if it were an increase. What results is the amount of the corrected Arc of Separation.* If it is equal to the (equatorial) arc between the sun and the moon at their time of setting, or less than this, you know that the moon should be visible without doubt, whether it were obstructed by some (of the things) we mentioned or not. If the corrected arc is larger than the time of the settings, you know that the crescent can not be seen in that country.

Having expressed what results for the fraction in equator time, we determine (from a table) the times which we described as the times of rising of the degree opposite the lunar degree in the climate. The ecliptic degree corresponding with it, is the degree opposite the degree with which the moon sets. We add to it what we obtained for the fraction, and keep the result. This we do if the fraction were an increase. If it were a decrease, we subtract it from the ecliptic degrees which we obtained. (Again) we keep what results. For (one of) the two situations, whichever suits us, we find the corresponding rising times. We see how much more or less the amount is than the first times, which are the times of rising of the degree opposite the lunar degree. What results is the amount of the resulting fraction expressed in equator time. We subtract it from the Arc of Separation, or add it to it, as required, by God's will.

(5. Last crescent of the rising moon)

The disappearance of the moon in the morning at the ends of the months (can be determined) in the same way. (Now), however, you should use the times of rising of the solar degree itself (and not, as before, the times of the degree opposite to it). The degree of rising of the moon is found by subtracting half the arc of the lunar day from the rising times of the equator degree at mid-sky. What remains is the rising time of the degree in which the moon appears in that climate. From this you subtract the rising time of the solar degree. What remains is the amount between the rising times of the sun and the moon, when the moon is in the East.

If the Arc of Separation which results is equal to the arc between the rising of the sun and the moon, or less, the moon may be seen in the morning before sunrise at the time of sunrise. If it is more, the moon remains hidden in the sun's light, and will not be seen. (In this case) a table for the sun and the moon should be constructed for the time of sunrise on the 28th day of the Arabic month, which precedes the conjunction of the moon and the sun by one day.

(6. The shape of the crescent and the direction of its horns)

(If you wish to know) the figure of the crescent as its appears, whether its horns are level or tilted, and the amount of light it contains, divide the (angular) distance between the sun and the moon, marked off on the ecliptic by the latitude of the moon, into parts of the 12 digits of the moon's disk. What results are the digits of the (Arc of) Light. Now draw a circle (for the solar disk) of arbitrary size, and intersect it by two lines crossing at right angles in the centre (Figs. 15a, b). At the ends of these lines place the (cardinal) directions of the horizon. Divide each quadrant of the circle into 90 degrees and mark off

*Al-Battani has Arc of Vision. As he assumes the sun and moon to be on the equator, the Arc of Light and the Arc of Separation coincide, and he calls them Arc of Vision. It is not clear, however, why the elongation also should be identified as Arc of Vision

the latitude of the true moon from the East-point and the West-point in the direction of the moon's latitude, so that the amount of each arc is equal to the moon's latitude. Then place the edge of the ruler on the two marks and draw a line passing through them parallel to the diameter of the circle. Let it extend outside the circumference towards the East by an amount equal to the radius of the circle. Along this line the moon will proceed in longitude at that particular time. At other times it will (proceed) according to its latitude. At the time when its light reaches to the middle (first quarter), the centre of this disk lies at the inter section of the circumference of the circle and this line. And from the time when it is half filled with light to the time when it is full, the centre of the circle (of the moon) will be on the line extending outside the circumference of the circle towards the East, where it reaches the end of the line (which lies) between its disk and the circle of the sun. (If the lunar latitude is zero) the first circle which is drawn for the sun becomes the moon's disk when (the moon) is new.

Now mark off on the circumference of the circle, from the North-point towards the East, an amount equal to the distance between the sun and the moon. Make also a mark from the South towards the East. Connect these two marks by a line. Where the two lines intersect is the centre of the moon's disk. Draw a circle around it equal to the first circle. The segment which (now) falls between the two arcs has the appearance of the crescent (of the moon) and exactly the same shape. Now connect the two points in which the two circles intersect by a line. This will be (a chord parallel to) the second diameter of the circle. Then draw another straight line passing through the centres of both circles and through the two arcs. (This line will be at right angles to the chord), so that it cuts the crescent into two halves. By means of the degrees which you marked on the circumference (of the first circle) you will be able to find out how much each of the two horns of the crescent is tilted with respect to the ecliptic. The (inclination of) the ecliptic with respect to the horizon can namely be found from the direction of the (ecliptic) arc, which at that time rises or sets with (the moon).

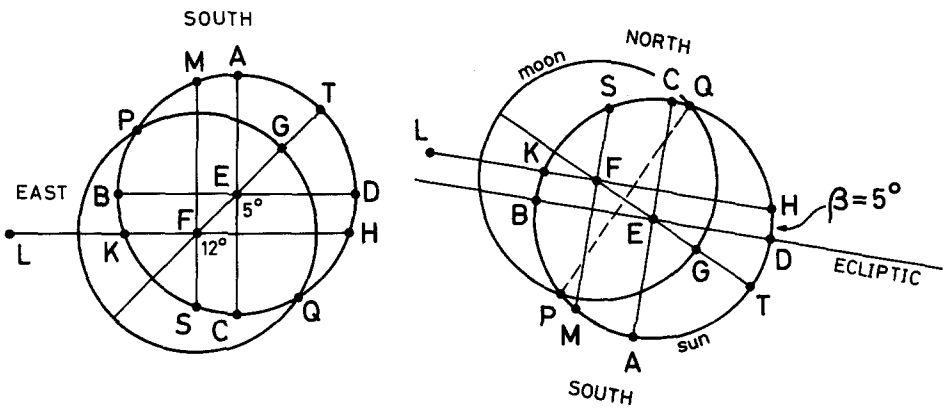


Fig. 15a, (and b.)

(We will now describe the above procedure more precisely.) Let the circle for the sun be ABCD, its centre be E, and two diameters AC and BD. Let A be in the South, C in the North, B in the East and D in the West. Let us assume that the moon's northern latitude is 5° and that its true distance from the sun is 12°. On the circle we take from points B and D (amounts)

equal to the moon's latitude towards the North, which is towards point C. These (points we call H and K and) we connect them by a line, which is HK. We extend it to L, and let line KL be equal to EB. We (also) take from A and C towards B two arcs, each of which is of an amount equal to (the distance) between the sun and the moon. We mark the ends of the two arcs M and S, and connect them by a line MS. The point where the line HK intersects (MS) we call F. We take this as a centre, and draw a circle for the moon around it, equal to the first circle. The points of intersection of the two circles we call P and Q. We also draw line EF and extend it to point T which bisects arc PQ. The point where line FET intersects the circumference of the moon's circle we call G. Then line TG bisects the curve of the crescent and the middle of the light (sickle) and is the amount in digits of the lunar light. The two points P and Q are the horns of the crescent and the inclination on the line of equality (TG) with respect to the ecliptic is given by arc AP and arc CQ. In this case D will be towards the western (part) and B towards the eastern (part) of the circle of the horizon, while BD will be a line (segment) of half the ecliptic.

By this diagram, therefore, you can know at any time of any month the shape of the light of the crescent according to the distance (of the moon). Whenever the moon is at its closest distance (from the earth) the horns (of its crescent will be shorter) because its disk will be larger than (the disk of) the sun.

And that is what we wished to show, by God's will.

(7. The actual observation of the crescent)

If you wish to fix the position where the crescent may be seen in its orbit at the beginnings of the months, according to its altitude above the western horizon as well as according to the azimuth of its altitude circle—which passes through the zenith, the moon and (a point on) the horizon—do as follows:

Add about 8 minutes of time to the degree which is at mid-sky with the moon, i.e. the degree which lies at mid-sky at the time of observation of the moon. (This is done) because the rays of the sun prevent observation from the time of sunset, until it has gone down beyond the horizon by an amount of approximately 1/8 th of an hour (or about 8 minutes).

Next determine the apparent altitude of the moon at about 1/8 th of a mean hour after sunset and determine the moon's azimuth on the circle of the horizon at that time, in the way we described.

Then go to a location with an unobstructed horizon and erect there a pole, or something that can be used as a pole, with a height above the ground equal to your own height, so that the observer can project its (top) onto the moon (Fig. 17). The surface (of the location) should be levelled and balanced with a plumb-line and (be made) parallel to the plane of the horizon. Choose a centre in it and draw a circle of any convenient size. Then draw the directions of East, West, South and North, as we mentioned (in Chapter 12) for the determination of the meridian line and divide the quadrant of the circle which is towards the side of the crescent into 90 degrees. Then take a leveled ruler, or a hollow reed, and place the edge of the ruler or the axis of the reed above the centre of the circle, in the direction of the azimuth of the crescent towards either the East or the West point, depending on which side the crescent is.

Next suspend an astrolabe from your hands with the alhidade at an angle equal to the altitude of the apparent moon, as you determined it. Then lift the end of the ruler or the reed which is towards the crescent above the surface of the circle from what supports it (by placing something under it) without letting it deviate from the azimuth of the moon or from the centre of the circle, in such a way that the end pointed at the moon will rise and the end where one observes will fall. The ray of vision has to pass through the two holes of the alhidade and along the ruler's edge, or (along) the axis of the reed. Thus it will fix a line from the position of the eye to the lunar position at that azimuth.

If the observer looks at the time of appearance he will see the crescent in the direction of the ruler's edge or through the reed. And this is the diagram of what we have described, by God's will.

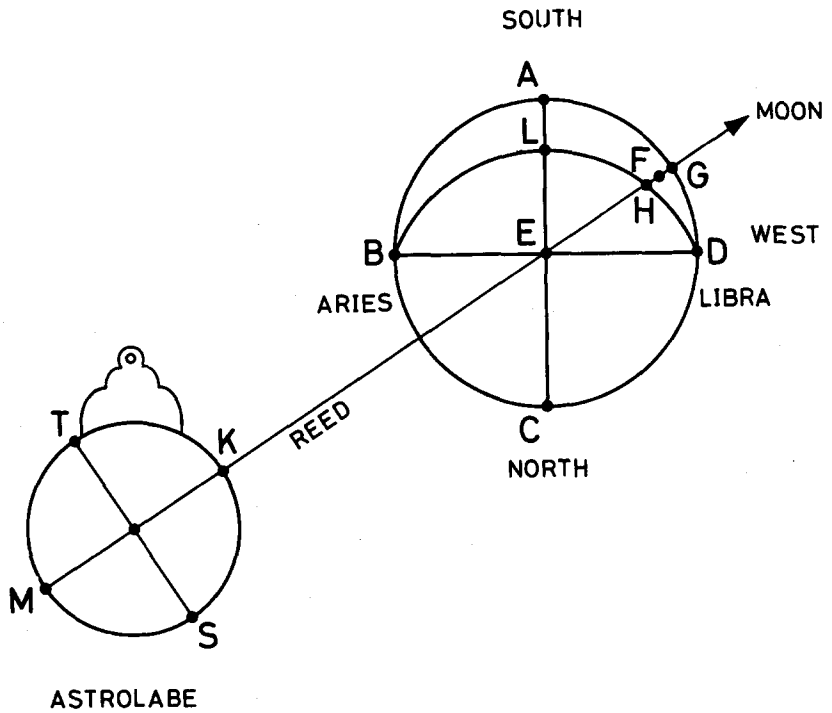
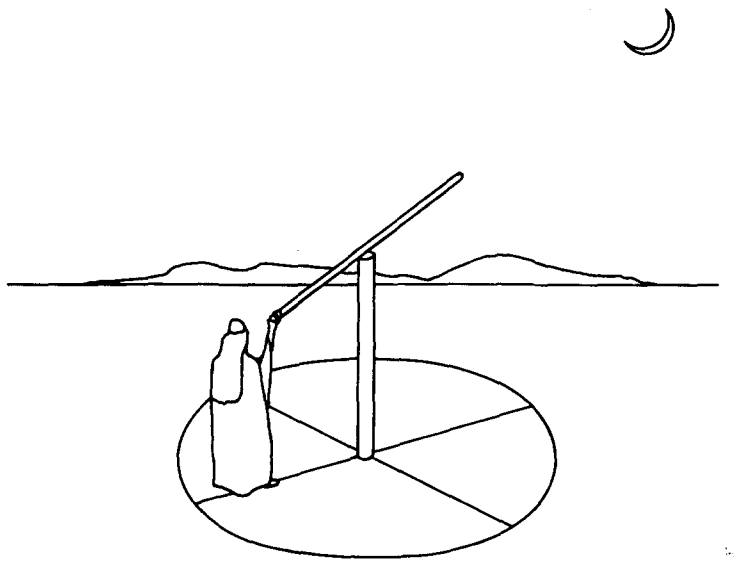


Fig. 16.

(We will now describe the above procedure more precisely.) Around centre E draw the circle of the horizon ABCD. Let E be the position of the centre of the horizon on the flagstone. A is the South-point, B the East-point, C the North-point, D the West-point. Then draw the two lines AC and BD. We assume the moon to be in the West, which is in the quadrant AD. (For simplicity) we take point B in the ecliptic at the beginning of Aries, so that point D becomes the beginning of Libra. These (then) are the (degrees) of rising and setting of the ecliptic. We assume arc DLB to be the southern half of the ecliptic. It is then clear that the beginning of Capricorn lies on line CL, which is on the prime meridian. The degree which lies as mid-sky with the moon will be point H of the ecliptic, and this is the beginning of Scorpio. The position of the moon, having southern latitude, we assume to be F. We draw line KEHFG which is the position of the ruler's edge or the axis of the reed, which passes through the centre of the circle onto the position of the moon (F) and the degree (H) which is at mid-sky with it. We now determine the arc FG above the horizon. It is clear that arc FG is the altitude of the moon. AL is the altitude of the beginning of Capricorn at mid-sky,

and arc DH is the arc of the ecliptic from the point of the beginning of Libra to the degree at mid-sky with the moon. Point G gives the azimuth of the moon. Thus arc DG of the horizon is the distance of the direction of the moon from the equinoctial West-point.

So, when line KG is raised from point E and point G towards the sky as much as the moon's altitude, (as) marked off on the astrolabe, point K will be lowered towards the earth. The (ray of) vision will pass through the two holes of the alhidade of the astrolabe. These are points M and K. The whole is joined to one line MG. So, if the observer looks from position M or position K he may see the crescent with that device along the direction of line KG. If the air is pure and thin, this is beyond doubt, but if the weather in that country is changing and bad, it may obstruct its observation. It may (then) be seen in countries other than this one, as long as it has the same distance in equator degrees as in that country. It is not necessarily so that the sky is overcast all over the country. So, (although) it cannot be seen in your place, it may be that (it is seen) from nearby villages or homes.



(Fig. 17.)

Al-Battani's Theory on the Appearance of the Crescent in the Evening

A commentary by G. V. Schiaparelli
(translated from the Latin)
Milano, 1903

In addition to a clear sky, the author mentions three principal conditions under which the lunar sickle may be seen on the evening of the new crescent:

1. The angular distance from the moon to the sun (the Arc of Light)

This is a measure of the visible sickle. Let O be the observer (Fig. 18) on the surface of the earth, OS the direction of the sun, LS the line from the moon to the sun. Because of the very large distance of the sun, OS and LS can be considered parallel. CED is the illuminated hemisphere, which is determined by plane CD, perpendicular to LS and OS. AFB is the hemisphere determined by plane AB perpendicular to OL, which can be seen by the observer. The part of the moon which is seen in the shape of a sickle is the segment CLA, of which the amplitude CA

(the Arc of Light) can be given in degrees. Angle CLA, namely, equals angle SOL, the angle formed by the centres of the sun and the moon at the eye of the observer. Al-Battani states that, if the observations take place under average conditions (with the sun and moon on the equator), the lunar crescent can only be seen for the first time when angle SOL equals 12 degrees. He adds that these 12 degrees do not differ much from the mean (ecliptic) distance which the moon moves away from the sun in one day, which is $12^{\circ}11'$. We may call this the *Mean Elongation*. This is in approximate agreement with those who hold that the moon can not be seen less than 24 hours after true conjunction. As a basis of his calculation he chooses $12^{\circ}11'$.

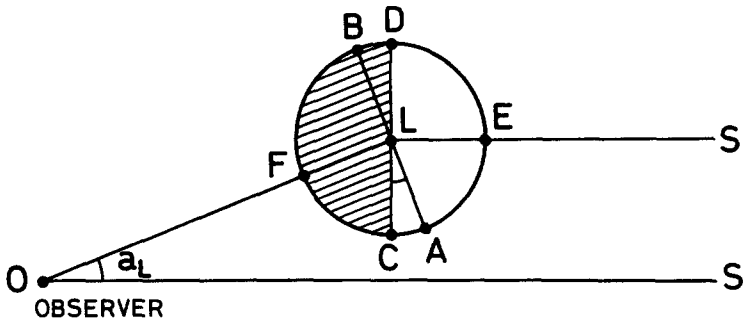


Fig. 18.

2. The brightness of the lower part of the sky where the moon is in the evening

This brightness is determined by the dip of the sun below the horizon at that moment. When the moon and the sun are on the equator (which is normally not the case) it appears from the above that a sufficient dip can be reached if it takes from sunset to moonset a time interval corresponding to an equatorial distance of 12 degrees, or 4/5 th of an hour. In this 4/5 th of an hour the moon moves away from the sun by 2/5 of a degree, or 24 minutes of arc. If it is assumed that the elongation between the two luminaries is to be $12^{\circ}11'$ at the moment of moonset, it will be 24' less at sunset, or $11^{\circ}47'$. Al-Battani rounds this off to $11^{\circ}45'$. However, if the line connecting the two luminaries is not parallel to the equator, the (ecliptic) arc of $12^{\circ}11'$ will not constitute the limit of appearance. In the region of the town of Raqqah (where al-Battani lived) sometimes this line may run almost parallel to the horizon, because of the inclination of the ecliptic and the latitude of the moon. And at other times it may be at right angles with the horizon and the brightness of the sky. Therefore al-Battani posits instead, as a general rule, that the vertical (Arc of Descent) between the solar parallel and the moon be passed beyond the horizon in 4/5 th of an hour. This implies that in the average situation there should be 12 equator degrees, or 4/5 th of an hour, between the two settings, corresponding—as explained—to a Mean Elongation of $11^{\circ}45'$ at the time of sunset. In this case the dip of the sun below the horizon will be about constant. In the region of Raqqah (the Arc of Descent) may vary between about 10° and 8°, depending on the season, (whereas the solar dip at which the sickle becomes visible lies between 2 and 4 degrees).

3. The distance of the moon from the earth

Al-Battani remarks that the appearance of the moon also depends on the width of the luminous sickle. Under equal conditions this width is proportional to the radial parallax of the moon, and follows the same laws of variation. In the computation of these variations he considers the changes in the distance of the moon according to its position in the epicycle. He assumes

that one has the average situation when its distance equals the distance of the centre of the epicycle from the earth. The 60 th part of the correction of the anomaly due to parallax for 30' is found in the third column of table T II. He also assumes that the fixed amount of 12°, or 4/5 th of an hour (mentioned above), corresponds to the mean distance of the moon.

Having assumed all of this, al-Battani argues as follows: The conditions under which the moon is visible are determined by three variable factors, of which the increase of one may compensate the decrease of another. If, for instance, the angular distance between the moon and the sun is larger, the Arc of Vision will be proportionally wider. Therefore, in order for the moon to be just visible, a smaller dip of the sun will be allowed, and because of the proportionality the mean Arc of Vision ($11^{\circ}45'$) can be smaller. Also, if the moon is closer to the earth than the mean distance, its apparent size will increase proportionally and again a smaller dip of the sun will be required for the moon to appear. Finally, if at the same time the elongation and its distance are larger, the corrections to be made for the mean Arc of Vision must be combined. From this al-Battani derives his rules, which can be set forth as follows:

1. For the given day, at the moment of sunset, one derives from the tables the positions of the sun and the moon, taking into account the parallax. One determines the difference in time between moonset and sunset for the positions thus computed. Let a_s be this difference expressed in equator degrees. If now the moon shows its average sickle corresponding to a distance of $12^{\circ}11'$, and has its mean distance to the earth the problem is solved by comparing a_s to the mean Arc of Vision of $11^{\circ}45'$. In this case the moon would be visible that evening if $a_s \geq 11^{\circ}45'$.
2. If at that moment the elongation of the moon from the sun differs from $12^{\circ}11'$, one should compute the angular distance as it appears to an observer on earth. (Taking into account the parallax as well as its fraction of) the mean elongation of $12^{\circ}11'$. If the fraction is larger than unity, the arc should be decreased proportionally. The result is the corrected Arc of Vision $\bar{a}_v$, adjusted to the mean elongation of $12^{\circ}11'$. Indeed, if the actual elongation is larger than the mean elongation, the lunar sickle must be proportionally wider, so that the moon becomes visible for a smaller Arc of Vision. This arc, adjusted in this way, is the one to be compared with a_s, if the moon had the mean distance to the sun.
3. If the moon does not have the mean distance, one should proceed, by means of the corrected anomaly of the moon, to the third column of table T II (p. 89) which serves for the analogous reduction of the parallax. The radial parallax and the amplitude of the observed lunar sickle decrease in the same proportion as the distance. The mean sickle corresponds with number 30 of column 3, the smallest with zero, the largest with number 60. A variation of 30 parts corresponds to 1/12 th in the observed size of the sickle. If the number in the third column, corresponding with the given anomaly, is called n, and if $n < 30$, the sickle will become smaller according to the ratio

$$1 \; : \; (1 - \frac{1}{12} \; \frac{30 - n}{30}) \quad .$$

If $n > 30$, the sickle will increase by

$$1 \; : \; (1 + \frac{1}{12} \; \frac{n - 30}{30}) \quad .$$

If the sickle has become larger (i.e. when $n > 30$), the Arc of Vision a_v, as computed above, must become smaller in the same proportion. If the arc has become smaller ($n < 30$), arc a_v must become larger according to the inverse calculation. The result is amount $\bar{\bar{a}}_v$ which is the arc corrected for both variations, namely the one of elongation and the one of distance to the earth.

Finally arc a_v has to be compared to $\bar{\bar{a}}_v$. If a_v is larger, and the sky is clear, the moon will be visible. If a_v is smaller, one has to wait until the following day.

(4. Limits of the Arc of Vision)

Al-Battani now discusses a different way to carry out the same computation. He bases himself on the same principles, but takes different numbers from which to start. He assumes that, when the moon is at its largest distance from the earth, i.e. when it has the smallest diameter as seen from the earth, it will become visibile when it has an Arc of Vision of $10^{\circ}51'$ and an angular distance from the sun of $13^{\circ}40'$. He explains in a somewhat strange fashion the choice of these amounts by saying that the first speed is the smallest possible speed of the sun with respect to the moon in the ecliptic—i.e., of course, when the sun has the largest hourly motion and the moon has the smallest—and the second is the largest relative speed, when the sun moves as slowly as possible and the moon as fast as possible. But, however it may be, as a result of these amounts the situation does not differ very much from the conditions chosen in the previous section. According to what he postulated before one finds that

$$11^{\circ}45' \cdot \frac{13}{12} \cdot \frac{12^{\circ}11'}{13^{\circ}40'} = 11^{\circ}21' = \text{Arc of Vision},$$

corresponding with the largest lunar distance and an elongation of $13^{\circ}40'$. Now al-Battani takes only $10^{\circ}51'$ instead of $11^{\circ}21'$. The difference is of little importance, but nevertheless—according to the new computation—the new moon may now be found to be visible under conditions where it would not have been visible previously. The manner of computation, however, is the same as in the former case, except that for the computation of the distance of the moon from the earth the largest distance instead of the mean distance is taken as a basis. After this one proceeds as before.

(6. The shape of the crescent and the direction of its horns)

The construction of the author gives the inclination of the horns with respect to the ecliptic with sufficient accuracy but, of course, only as long as the elongation between the sun and the moon is not much larger than it usually is on the day of tis first appearance. One should also pay attention to the fact that GT is the true width of the visible lunar sector, not the width of the sickle. The latter is much smaller than the sector, because the lunar surface is seen under a very inclined angle. In this way we have to understand "the shape of the light of the crescent".

(7. The actual observation of the crescent)

In Fig. 16 there are some things which do not agree with the text. (The figure is repeated below as Fig. 19). In the text some statements are not correct.* Since EG is the vertical circle which passes through the moon F and marks its azimuth G on the horizon, point H of the ecliptic, which culminates with the moon, should not be on the line passing from G to E, but on the large circle which passes from the moon F to the celestial pole P (as shown here in Fig. 18). Yet in the texts of the "Codex" and of "Plato" the line is called KEHFG. The figure is poorly described and the text adapted to the errors of it, as we see it elsewhere. I think that the original figure of al-Battani has been as I describe it here. But then we must reject what has been printed in cursive on page 91 (of Nallino's translation). Point H itself is useless. Why the moon should be in the South I do not know, nor why Libra should be in D. Still less do I understand why H should be the beginning of Scorpio.

*Schiaparelli's objections seem to be partly due to Nallino's Latin translation, which is not clear here.

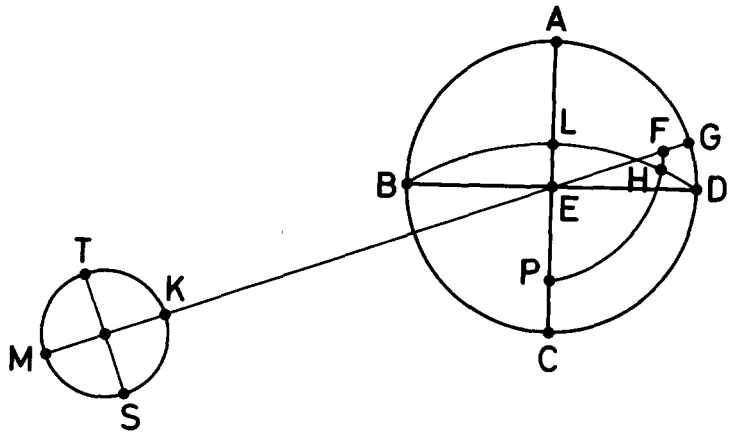


Fig. 19.

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